

Automated email audit

Every email the system sends without a human, exactly as the recipient sees it, and exactly when it fires.

20

MEMBER EMAILS
FIRE AUTOMATICALLY

6

OPERATOR ALERTS
FIRE AUTOMATICALLY

5

BUILT BUT
NEVER AUTO-SEND

43

RENDERED BODIES
IN THIS DOCUMENT

Every automated send, and when it fires

20 member-facing emails and 6 operator alerts fire with no human in the loop. A further 5 exist in the codebase but are **not** wired to send on their own — those are listed separately, and the distinction matters.

REACHES A MEMBER'S INBOX AUTOMATICALLY

EMAIL	FIRED BY	EXACTLY WHEN
Email verification	APP ACTION	Immediately on sign-up, when the account row is created.
Password reset	APP ACTION	Immediately on a "forgot password" request.
Verify reminder	SYSTEMD TIMER	2–168 hours after registering, if the address was never confirmed. Swept every 2 hours.
Trial / paid welcome	STRIPE WEBHOOK	Seconds after checkout completes — as soon as Stripe posts the event.
Founding-member welcome	STRIPE WEBHOOK	Seconds after a founding-tier checkout completes.
Welcome back (resubscribe)	STRIPE WEBHOOK	Seconds after a previously-lapsed account subscribes again.
Mid-trial value nudge	SYSTEMD TIMER	120 hours (5 days) BEFORE trial end — i.e. ~day 2 of a 7-day trial. ±3h window, swept every 6 hours.
48h trial-end reminder	SYSTEMD TIMER	48 hours BEFORE the trial converts. ±3h window, swept every 6 hours.
Trial conversion confirmation	STRIPE WEBHOOK	Seconds after the trial's first real charge clears.
Trial conversion declined	STRIPE WEBHOOK	Seconds after the FIRST failed attempt on the conversion charge. Silent on Stripe Smart Retries.
Payment failed	STRIPE WEBHOOK	Seconds after the FIRST failed renewal attempt. Retries send nothing.
Grace-expiry warning	SYSTEMD TIMER	≤24h before the payment-recovery grace window closes AND ≥12h after it opened. Swept every 4 hours.
Payment recovered	STRIPE WEBHOOK	Seconds after a failed payment finally clears.
Card expiring	SYSTEMD TIMER	When the card on file expires within 45 days and has not already been warned this calendar month. Checked daily.
Referral reward	APP EVENT	At the moment the referral is credited (the referee's subscription qualifies).
Abandoned-checkout recovery	SYSTEMD TIMER	24–168 hours after a billing_checkout_started audit row with no subscription to show for it. Swept every 6 hours.
Verified, never paid	SYSTEMD TIMER	2–168 hours after verifying, if checkout was never opened. Swept every 2 hours.
Reactivation (extended trial)	SYSTEMD TIMER	21+ days after signing up, for a verified account that never started a trial. Checked daily. Max 50 per run.
Cancellation acknowledgment	STRIPE WEBHOOK	Seconds after the member clicks Cancel — while they still have access through period end.
TradeWorkz bot alert	QUEUE DRAIN	Within ~60 seconds of the bot event. The timer drains the queue every minute.

REACHES THE FOUNDER INBOX AUTOMATICALLY

EMAIL	FIRED BY	EXACTLY WHEN
Cancellation alert	SYSTEMD TIMER	Within 15 minutes of a churn event. Swept at :07, :22, :37, :52 past every hour.
Win-back review digest	SYSTEMD TIMER	Weekly, Mondays 16:35 UTC.
Return-intent review digest	SYSTEMD TIMER	Daily at 16:50 UTC.
Scheduled-unit failure alert	SYSTEMD ONFAILURE	The instant any scheduled unit exits non-zero. Not a schedule — a failure hook.
Signup-rate alarm	SYSTEMD TIMER	Hourly at :35 UTC. The script itself gates to active hours and only alerts when the signup rate drops below the floor.
Folds of Honor donation reminder	SYSTEMD TIMER	Quarterly — the 5th of January, April, July and October at 09:00 ET.

BUILT, BUT NOTHING FIRES IT ON ITS OWN

EMAIL	STATUS	WHY IT DOES NOT SEND
Trial quickstart bridge	MANUAL	Never automatically. One-shot backfill, run by hand.
Founding final call	MANUAL	Never automatically. Run by hand the day before the founding deadline.
Win-back	SYSTEMD TIMER (DIGEST ONLY)	The timer runs weekly, Mondays 16:35 UTC — but it only mails the OPERATOR a review digest. No member is mailed until someone runs make winback YES=1 by hand.
Return intent	SYSTEMD TIMER (DIGEST ONLY)	The timer runs daily at 16:50 UTC — but it only mails the OPERATOR a review digest. No member is mailed until someone runs make return-intent YES=1 by hand.
Reactivation review digest	MANUAL	Never automatically. --digest is an exclusive branch that exits before sending, and the unit does not pass it.

Where each email lands in a member's life

Read top to bottom. Every box is one email; the grey ones need a human to press send.

● Sign up

Email verification
instant

Verify reminder
+2h if unconfirmed

● Verified, no checkout

Verified, never paid
+2h

Reactivation (30-day trial)
+21 days

● Started checkout, stopped

Abandoned-checkout recovery
+24h

● Trial starts

Trial / paid welcome
instant

Founding welcome
instant (founding)

● Mid-trial

Mid-trial value nudge
120h before end

● Trial ends

48h trial-end reminder
48h before

Trial conversion confirmation
on charge

Trial conversion declined
on decline

● Subscribed

Card expiring
45 days out

Referral reward
on referral

TradeWorkz bot alert
per bot event

● Payment fails

Payment failed
attempt 1

Grace-expiry warning
24h before cutoff

Payment recovered
on clear

● Cancels

Cancellation acknowledgment
instant

Cancellation alert → founder
within 15 min

● Lapsed

Win-back
DIGEST ONLY at 30 days

Return intent
DIGEST ONLY on their return

● Comes back

Welcome back
instant

The clock

Every systemd timer that can send mail, fastest first. Times are **server time (UTC)** except where marked ET. Each unit carries a randomised delay (15s–10m) so they do not all fire on the same second.

UNIT	ONCALENDAR	CADENCE	SENDS
zerogex-web-tradeworkz-notify	*:0/1:12	every minute	TradeWorkz bot alert
zerogex-web-cancellation-alerts	*:07,22,37,52	every 15 min	Cancellation alert (operator)
zerogex-web-signup-alarm	*:35:00	hourly	Signup-rate alarm (operator)
zerogex-web-verify-reminders	00/2:20:00	every 2h	Verify reminder
zerogex-web-verified-never-paid	00/2:30:00	every 2h	Verified, never paid
zerogex-web-grace-expiry-warnings	00/4:35:00	every 4h	Grace-expiry warning
zerogex-web-trial-reminders	00/6:15:00	every 6h	48h trial-end reminder
zerogex-web-checkout-recovery	00/6:45:00	every 6h	Abandoned-checkout recovery
zerogex-web-trial-value-nudge	00/6:45:00	every 6h	Mid-trial value nudge
zerogex-web-card-expiry	04:20:00	daily	Card expiring
zerogex-web-reactivation	16:40:00	daily	Reactivation — sends to members
zerogex-web-return-intent	16:50:00	daily	Return-intent digest — operator only
zerogex-web-winback	Mon 16:35:00	weekly	Win-back digest — operator only
zerogex-web-foh-donation-reminder	Jan/Apr/Jul/Oct 5th 09:00 ET	quarterly	Folds of Honor reminder (operator)

Also automatic, but not on a clock: the scheduled-unit failure alert fires from OnFailure=zerogex-web-alert@%n.service the instant any unit above exits non-zero — that is the email that tells you the rest stopped working.

ACCOUNT & AUTH · 2 EMAILS

Fires on an action the person just took. No opt-out — these are transactional.

- APP ACTION
- MEMBER INBOX

Email verification

WHEN Immediately on sign-up, when the account row is created.

FIRED BY	core/serverAuth.ts — on registration
CADENCE	Once per registration
WHO GETS IT	Anyone who registers.
SENT-ONCE GUARD	None — reissued whenever a new verification link is minted.
OPT-OUT	None (transactional)
SOURCE	core/mailer.ts:279 · core/serverAuth.ts
NOTES	Link expires in 24h. Verification gates checkout, so this is on the critical path to revenue.

FROM

ZeroGEX <hello@zerogex.io>

SUBJECT

Verify your ZeroGEX email

Verify your ZeroGEX email

Welcome to ZeroGEX! Please confirm this email address to finish setting up your account. Verification is required before you can subscribe to a paid plan.

Verify email

Or copy this URL into your browser:
<https://zerogex.io/verify?token=abc123def456>

This link expires in 24 hours.

If you didn't create a ZeroGEX account, you can safely ignore this email.

Password reset

WHEN Immediately on a "forgot password" request.

FIRE BY	app/api/auth/password/forgot/route.ts
CADENCE	Once per request
WHO GETS IT	Anyone who submits the forgot-password form.
SENT-ONCE GUARD	None — rate limiting lives in the route, not a DB latch.
OPT-OUT	None (transactional)
SOURCE	core/mailer.ts:2860
NOTES	Link expires in 30 minutes, single use.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Reset your ZeroGEX password**

Reset your ZeroGEX password

Someone requested a password reset for your ZeroGEX account.

If that was you, click the button below to set a new password:

Reset password

Or copy this URL into your browser:

<https://zerogex.io/reset?token=abc123>

This link expires in 30 minutes and can only be used once.

If you didn't request this, you can safely ignore this email — your password will not change.

WELCOME & ONBOARDING · 4 EMAILS

Fires once, off the Stripe subscription lifecycle.

STRIPE WEBHOOK

MEMBER INBOX

FOLDS OF HONOR FOOTER

Trial / paid welcome

WHEN Seconds after checkout completes — as soon as Stripe posts the event.

FIRED BY	checkout.session.completed / customer.subscription.created → maybeSendPaidWelcomeEmail
CADENCE	Once per account (first paid checkout)
WHO GETS IT	First subscription on the account; not a founding-tier checkout (that branches to the founding welcome).
SENT-ONCE GUARD	paid_welcome_email_sent_at — never cleared
OPT-OUT	None (transactional)
SOURCE	core/mailer.ts:374 · app/api/webhooks/stripe/route.ts:1233
NOTES	trialDays is measured off the subscription's own trial_start → trial_end, not a constant, so a 30-day reactivation trial renders "30-day free trial".

FROMZeroGEX <hello@zerogex.io>

SUBJECTYour ZeroGEX trial is active

VARIANTtrial (7-day)

Hello,

Your 7-day free trial is now active, so you have full access right away — dive in and make the most of it. You won't be charged until September 18, 2026, and if ZeroGEX turns out not to be the right fit, you're free to cancel before then from the billing portal on your [account page](#) and you won't be billed.

I just wanted to personally thank you for trying ZeroGEX.

The best place to start is the live dashboard. For your first session, here's where I'd look first:

- **Today's Read** — the plain-English summary at the top of the dashboard telling you whether conditions currently favor stability or bigger moves.
- **GEX Strike Profile** — the at-a-glance chart of how much dealer gamma sits at each strike, with the key walls and flip marked right on price, so the levels that matter jump out and are easy to act on.
- **Gamma Flip** — the line between more stabilizing and more volatile dealer positioning.
- **Call Wall / Put Wall** — the key upside and downside options levels where price may pin, reject, or react.
- **Net GEX** — whether the broader options structure is more stabilizing or more unstable right now.

- **SPY / SPX / QQQ / NDX / ES / NQ** — the same read across the cash indices and the ES / NQ futures, for intraday context wherever you trade it.

Open the live dashboard

ZeroGEX is built to give you better market-structure context before price gets there. It isn't financial advice, and no tool can promise a guaranteed outcome — how you trade the levels is always your call.

What it is, and what it isn't. Dealer positioning is *modeled*, not observed — no public dataset says who is long and who is short at a strike. So the levels are probabilistic context, not mechanical triggers. Exactly what's derived, what's assumed, and where the model is weakest is written down in [Methodology & Validation](#) — worth five minutes before you judge a level.

One more thing worth knowing about: your Pro plan includes self-service API keys, so you can call the ZeroGEX data API straight from your own scripts, spreadsheets, and integrations — no waiting on support. If you ever need one:

1. Open Account → API Access.
2. Click "Generate API Key" and copy the secret — it's shown only once.
3. Send it on your requests as the header: Authorization: Bearer <your key>

[Generate an API key](#)

You can generate or regenerate a key anytime from your account page — regenerating immediately deactivates the previous key.

ZeroGEX is still growing quickly, and early users like you help make it possible for me to keep improving the platform, adding features, and making the data more useful for active traders.

Please feel free to reply directly if you run into anything, have questions, or see something that could be improved. I read every message, and customer feedback is a huge part of how I'm shaping the product.

Thanks again — I really appreciate your support.

Best,
Michael
Founder, ZeroGEX

P.S. Curious what we've shipped lately? See the latest product updates at <https://zerogex.io/updates>.



ZeroGEX is a Folds of Honor Proud Supporter.

3% of every subscription funds educational scholarships for the families of fallen and disabled U.S. service members. [See /giving](#) for the running total.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Thank you for subscribing to ZeroGEX!**
VARIANT immediate paid (no trial)

Hello,

I just wanted to personally thank you for subscribing to ZeroGEX.

The best place to start is the live dashboard. For your first session, here's where I'd look first:

- **Today's Read** — the plain-English summary at the top of the dashboard telling you whether conditions currently favor stability or bigger moves.
- **GEX Strike Profile** — the at-a-glance chart of how much dealer gamma sits at each strike, with the key walls and flip marked right on price, so the levels that matter jump out and are easy to act on.
- **Gamma Flip** — the line between more stabilizing and more volatile dealer positioning.
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- **Net GEX** — whether the broader options structure is more stabilizing or more unstable right now.
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[Open the live dashboard](#)

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Founding-member welcome

WHEN Seconds after a founding-tier checkout completes.

FIRED BY	checkout.session.completed → maybeSendPaidWelcomeEmail (founding branch)
CADENCE	Once per account
WHO GETS IT	Founding-tier checkout only.
SENT-ONCE GUARD	paid_welcome_email_sent_at (shared with the paid welcome)
OPT-OUT	None (transactional)
SOURCE	core/mailer.ts:590 · app/api/webhooks/stripe/route.ts:1231

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Thank you for subscribing to ZeroGEX!**

Hello,

Your founding rate is locked in — but you won't be charged until September 18, 2026. Your first payment, at your founding rate, happens then. Cancel before that from the billing portal on your [account page](#) and you won't be billed.

I just wanted to personally thank you for subscribing to ZeroGEX.

It genuinely means a lot to have your support this early. ZeroGEX is still growing quickly, and early paid users like you help make it possible for me to keep improving the platform, adding features, and making the data more useful for active traders. As a Founding Member your rate is locked in for the first year, and the 25% lifetime discount applies automatically after that.

One more thing worth knowing about: your Pro plan includes self-service API keys, so you can call the ZeroGEX data API straight from your own scripts, spreadsheets, and integrations — no waiting on support. If you ever need one:

1. Open Account → API Access.
2. Click "Generate API Key" and copy the secret — it's shown only once.
3. Send it on your requests as the header: Authorization: Bearer <your key>

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Founder, ZeroGEX



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Welcome back (resubscribe)

WHEN Seconds after a previously-lapsed account subscribes again.

FIRE BY	subscription sync → subscription_lapsed 1 → 0
CADENCE	Once per lapse → return cycle (re-arms on each new lapse)
WHO GETS IT	subscription_lapsed = 1 at the moment the new subscription syncs active.
SENT-ONCE GUARD	subscription_lapsed flag, cleared to 0 when this fires
OPT-OUT	None (transactional)
SOURCE	core/mailer.ts:1315 · app/api/webhooks/stripe/route.ts:1271
NOTES	Carries the API-key walkthrough as a repair notice — dropping below Pro revoked every key the account held.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Welcome back to ZeroGEX!**

Hello,

I just wanted to personally thank you for coming back to ZeroGEX — it really does mean a lot to have you here again.

ZeroGEX has kept growing since you were last subscribed, and returning users like you help me keep building. Your full access has been restored, so you can jump straight back into the data.

One thing to know if you were using the API before: your old key was revoked when the subscription lapsed, so any script, spreadsheet, or NinjaTrader chart still pointed at it needs a fresh one. Generating a replacement takes a few seconds:

1. Open Account → API Access.
2. Click "Generate API Key" and copy the secret — it's shown only once.
3. Send it on your requests as the header: Authorization: Bearer <your key>

[Generate an API key](#)

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Trial quickstart bridge

WHEN Never automatically. One-shot backfill, run by hand.

Does not send on its own — no timer is wired to it.

FIRED BY	make send-trial-quickstart — no timer
CADENCE	Manual only
WHO GETS IT	Members who were mid-trial before the activation-focused welcome shipped.
SENT-ONCE GUARD	None — the operator picks the recipients
OPT-OUT	n/a
SOURCE	core/mailer.ts:510 · scripts/send-trial-quickstart.mts
NOTES	Included for completeness: it is wired to a Makefile target but NOT to any systemd timer.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Getting the most out of your ZeroGEX trial**

Hello,

I'm Michael, the founder of ZeroGEX. You're currently on your ZeroGEX free trial, with full access through **September 18, 2026**. I recently put together a short guide on where to focus first, and since you started your trial before it existed, I wanted to make sure you had it too rather than leaving you to dig for the important parts.

The best place to start is the live dashboard. Here's where I'd look first:

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- **Gamma Flip** — the line between more stabilizing and more volatile dealer positioning.
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[Open the live dashboard](#)

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If anything's unclear or you have a question, just reply to this email. I read every message myself, and it genuinely shapes what I build next.

Thanks for giving ZeroGEX a try.

Best,
Michael
Founder, ZeroGEX



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TRIAL LIFECYCLE · 4 EMAILS

The trial clock. Two sweeps and two webhook bookends.

SYSTEMD TIMER MEMBER INBOX

Mid-trial value nudge

WHEN 120 hours (5 days) BEFORE trial end — i.e. ~day 2 of a 7-day trial. ±3h window, swept every 6 hours.

FIRE BY	zerogex-web-trial-value-nudge.timer → make trial-value-nudge YES=1
CADENCE	Timer: 00/6:45 UTC (every 6h, +0–5m jitter)
WHO GETS IT	subscription_status='trialing' AND trial_midpoint_email_sent_at IS NULL AND cancel_at_period_end=0 AND marketing_unsubscribed_at IS NULL AND deleted_at IS NULL AND current_period_end within target ±3h
SENT-ONCE GUARD	trial_midpoint_email_sent_at — permanent one-shot
OPT-OUT	Honors marketing_unsubscribed_at; carries an unsubscribe link
SOURCE	core/mailer.ts:946 · scripts/send-trial-value-nudge.mts (TARGET_HOURS 120)
NOTES	Deliberately carries NO charge or cancel language — its whole job is time-to-value, ahead of the day 3–7 cancel wave.

FROMZeroGEX <hello@zerogex.io>

SUBJECTGetting a real read out of your ZeroGEX trial

Hello,

You're a couple of days into your ZeroGEX trial, and I wanted to reach out while you've still got runway (your trial runs through **September 18, 2026**). Almost everyone who sticks around got one clear read early — so here's the fastest path to it:

1. Start on the [Main Dashboard](#) — the page to open every morning: your at-a-glance read of the regime, the key levels, and where price sits inside them.
2. Pull up the [Gamma Chart](#) — SPY/QQQ/SPX/NDX price with the **Gamma Flip**, **Call/Put Walls**, and **Max Pain** drawn right on it: the support/resistance map dealers actually defend. Session rewind lets you replay how a level held.
3. Check [Trade Bias](#) for a single, signed directional call — it fuses the gamma and volatility regime with live flow, tape, and momentum into one read, for a multi-day swing or a same-day ODTE.

Open your dashboard

Want the model behind it? The 5-minute read is [Gamma Exposure Explained](#).

And if you want to know what the model *doesn't* claim — what's measured, what's assumed, and where it's weakest — that's [Methodology & Validation](#). Dealer positioning

is modeled, not observed, and the levels are probabilistic context rather than mechanical triggers. I'd rather you read that and push back than guess at what the numbers are promising.

If anything's confusing, not what you expected, or just not clicking yet — reply to this email and tell me. I read every message and I'll personally help you get a useful read before your trial is up.

Best,
Michael
Founder, ZeroGEX

Prefer fewer emails like this? [Unsubscribe](#).

48h trial-end reminder

WHEN 48 hours BEFORE the trial converts. $\pm 3h$ window, swept every 6 hours.

FIRE BY	zerogex-web-trial-reminders.timer → make trial-reminders YES=1
CADENCE	Timer: 00/6:15 UTC (every 6h, +0–5m jitter)
WHO GETS IT	subscription_status='trialing' AND trial_reminder_email_sent_at IS NULL AND cancel_at_period_end=0 AND current_period_end within target $\pm 3h$
SENT-ONCE GUARD	trial_reminder_email_sent_at — re-arms on each fresh trialing window
OPT-OUT	None (billing notice — sends regardless of marketing opt-out)
SOURCE	core/mailer.ts:719 · scripts/send-trial-reminders.mts (TARGET_HOURS 48)
NOTES	The charge amount and card are resolved live from Stripe per member at send time. Anyone who already clicked Cancel is excluded — their trial will not convert. Carries NO discount: the 25% is a win-back lever, not something handed to a trialer who is about to convert anyway.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Your ZeroGEX free trial ends in 2 days**
VARIANT engaged

Hello,

A quick heads-up: your ZeroGEX free trial ends on **September 18, 2026 at 7:01 PM EDT** and **turns into a paid subscription automatically** — your first payment will be charged then unless you cancel before that.

Your subscription will begin at **\$59.00/month** using your Visa card ending in **4242**. Please make sure your payment method is ready, or [update it here](#).

If ZeroGEX is working for you, there's nothing you need to do — your access carries straight on and the subscription renews by itself.

If it isn't the right fit, you can cancel anytime before September 18, 2026 at 7:01 PM EDT from the billing portal on your [account page](#) and you won't be charged a cent.

[Manage subscription](#)

Either way, thanks for giving ZeroGEX a try — if there's anything I can do to make it more useful for you, just reply to this email. I read every message.

Best,
Michael
Founder, ZeroGEX

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Your ZeroGEX free trial ends in 2 days**
VARIANT on a promo intro rate

Hello,

A quick heads-up: your ZeroGEX free trial ends on **September 18, 2026 at 7:01 PM EDT** and **turns into a paid subscription automatically** — your first payment will be charged then unless you cancel before that.

Your subscription will begin at **\$29.00/month** using your Visa card ending in **4242**. Please make sure your payment method is ready, or [update it here](#).

Good news on the price: you locked in our **limited-time introductory rate** for the first 6 months, so that's what your card will be charged after the trial — not the standard rate.

If ZeroGEX is working for you, there's nothing you need to do — your access carries straight on and the subscription renews by itself.

If it isn't the right fit, you can cancel anytime before September 18, 2026 at 7:01 PM EDT from the billing portal on your [account page](#) and you won't be charged a cent.

[Manage subscription](#)

Either way, thanks for giving ZeroGEX a try — if there's anything I can do to make it more useful for you, just reply to this email. I read every message.

Best,
Michael
Founder, ZeroGEX

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Before your ZeroGEX trial converts — nothing you need to do**
VARIANT dormant (never returned)

Hello,

A quick heads-up so nothing catches you out: your ZeroGEX free trial ends on **September 18, 2026 at 7:01 PM EDT** and **turns into a paid subscription automatically**. There's nothing you need to do — your access simply carries on.

Your subscription will begin at **\$59.00/month** using your Visa card ending in **4242**.

If you haven't had a chance to dig in yet, that's the part I'd like to fix. Reply to this email and tell me what you trade — I'll point you at the two or three levels on the board that actually matter for it. That's usually the whole gap between signing up and it being useful.

And if you've decided ZeroGEX isn't for you, you can cancel your subscription from the billing portal on your [account page](#) before September 18, 2026 at 7:01 PM EDT and you won't be charged.

Either way, thanks for giving ZeroGEX a try — if there's anything I can do to make it more useful for you, just reply to this email. I read every message.

Best,
Michael
Founder, ZeroGEX

Trial conversion confirmation

WHEN Seconds after the trial's first real charge clears.

FIRE BY	invoice.paid → maybeSendTrialConvertedEmail
CADENCE	Once per account, ever
WHO GETS IT	isTrialConversionInvoice: a subscription_cycle invoice created within ~2 days of trial_end.
SENT-ONCE GUARD	trial_converted_email_sent_at — never cleared
OPT-OUT	None (receipt)
SOURCE	core/mailer.ts:1853 · app/api/webhooks/stripe/route.ts:1504
NOTES	The conversion predicate is what keeps this off ordinary renewals — invoice.paid fires on every cycle.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Your ZeroGEX trial just became a full membership**
VARIANT charged

Hello,

Your free trial just wrapped up and the first payment went through — \$59.00 on your Visa card ending in 4242. You're now a full ZeroGEX member. Nothing changes on your end and there's nothing to do — your full access simply carries on uninterrupted.

Your next charge is on October 14, 2026, and it renews automatically from there until you cancel.

Your invoices, your card, and the cancel button all live on your [account page](#) — you're free to change or cancel any of it at any time.

Open the live dashboard

Thank you for backing ZeroGEX this early. Paid members are what make it possible for me to keep improving the platform, and I don't take that lightly.

If anything about this charge looks off, or you have a question about your plan, just reply to this email — I read every one and I'm happy to sort it out.

Best,
Michael
Founder, ZeroGEX

ZeroGEX is a Folds of Honor Proud Supporter.



3% of every subscription funds educational scholarships for the families of fallen and disabled U.S. service members. [See /giving](#) for the running total.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Your ZeroGEX trial just became a full membership**
VARIANT fully credited (\$0 invoice)

Hello,

Your free trial just wrapped up and your membership rolled straight on — a credit on your account covered this first period in full, so there was nothing to pay. Nothing changes on your end and there's nothing to do — your full access simply carries on uninterrupted.

Your next charge is on October 14, 2026, and it renews automatically from there until you cancel.

Your invoices, your card, and the cancel button all live on your [account page](#) — you're free to change or cancel any of it at any time.

Open the live dashboard

Thank you for backing ZeroGEX this early. Paid members are what make it possible for me to keep improving the platform, and I don't take that lightly.

If anything about this charge looks off, or you have a question about your plan, just reply to this email — I read every one and I'm happy to sort it out.

Best,
Michael
Founder, ZeroGEX



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Trial conversion declined

WHEN Seconds after the FIRST failed attempt on the conversion charge. Silent on Stripe Smart Retries.

FIRE BY	invoice.payment_failed (attempt_count === 1) → trial branch
CADENCE	Once per failure episode
WHO GETS IT	Same isTrialConversionInvoice predicate as the success bookend.
SENT-ONCE GUARD	invoice.attempt_count === 1 gate (not a DB latch)
OPT-OUT	None (urgent billing)
SOURCE	core/mailer.ts:1574 · app/api/webhooks/stripe/route.ts:2318

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Your ZeroGEX trial ended — a quick card fix to keep your access**

Hello,

Your free trial just wrapped up and I went to start your subscription (\$59.00) — but your Visa card ending in 4242 was declined, so the first charge didn't go through. Good news: your full access stays on through September 19, 2026, so nothing changes right now. Update your card before then and you won't miss a beat — if a charge still can't be collected, the account moves to the free Public tier, and full access switches back on automatically the moment one succeeds.

Stripe will automatically try the card again on September 19, 2026, so an expired-or-replaced card or a momentary insufficient-funds hold may simply clear on its own.

You can update your card in about a minute from your [account page](#).

Update your card

If ZeroGEX earned a spot in your routine this week, that's all it takes to keep it. And if something's holding you back, just reply to this email — I read every one and I'm happy to help.

Best,
Michael
Founder, ZeroGEX

BILLING & DUNNING · 5 EMAILS

Payment failures, the grace window, and the recovery bookend.

STRIPE WEBHOOK MEMBER INBOX

Payment failed

WHEN Seconds after the FIRST failed renewal attempt. Retries send nothing.

FIRED BY	invoice.payment_failed (attempt_count === 1) → renewal branch
CADENCE	Once per failure episode
WHO GETS IT	An established (non-trial) renewal invoice that failed on attempt 1.
SENT-ONCE GUARD	invoice.attempt_count === 1 gate
OPT-OUT	None (urgent billing)
SOURCE	core/mailer.ts:1459 · app/api/webhooks/stripe/route.ts:2320
NOTES	The attempt_count gate is precisely why a second dunning touch exists — see the grace-expiry warning.

FROM

ZeroGEX <hello@zerogex.io>

SUBJECT

We couldn't process your ZeroGEX payment

Hello,

Your subscription payment of \$59.00 was declined — your Visa card ending in 4242 didn't go through. Your Pro access stays on through September 19, 2026, so nothing changes right now. If the payment still can't be collected by then, your account moves to the free Public tier — and full Pro switches back on automatically the moment a charge succeeds.

Stripe will automatically try again on September 19, 2026. If the retries don't succeed, the subscription will be canceled and you'd need to resubscribe to get Pro back.

If you'd rather fix it now, you can update your payment method in a minute from the billing portal on your [account page](#).

Update payment method

If it was just a temporary hiccup (insufficient funds, an expired or replaced card), the next retry may clear it with nothing needed from you.

And if you have any questions, just reply to this email — I'm happy to help.

Best,
Michael
Founder, ZeroGEX

Grace-expiry warning

WHEN ≤24h before the payment-recovery grace window closes AND ≥12h after it opened. Swept every 4 hours.

FIRE BY	zerogex-web-grace-expiry-warnings.timer → make grace-expiry-warnings YES=1
CADENCE	Timer: 00/4:35 UTC (every 4h, +0–5m jitter)
WHO GETS IT	subscription_status='past_due', grace window open, not already warned for THIS window, not cancelled, not deleted
SENT-ONCE GUARD	payment_grace_warning_sent_for — stores the window anchor, so a re-opened window re-arms it
OPT-OUT	None (urgent billing)
SOURCE	core/mailer.ts:1702 · core/graceExpiryWarning.ts (LEAD 24h / MIN_OPEN 12h)
NOTES	A sweeper, not a webhook: no Stripe event fires at "24h before the window closes". The --dry-run window-elapsed counter is an alarm — anything above zero is a member who dropped to Public with no heads-up.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Your ZeroGEX access ends September 19, 2026 — your last payment didn't go through**
VARIANT renewal failure

Hello,

Your most recent ZeroGEX payment on your Visa card ending in 4242 still hasn't gone through. I've kept your full access switched on while the card is retried — but that runs out on September 19, 2026.

Stripe will try again automatically on September 19, 2026, so an expired-or-replaced card or a momentary hold from your bank may still clear on its own.

If nothing clears by then, the account simply moves to the free Public tier. Nothing is deleted — your account, your settings and your history all stay exactly as they are, and full access switches back on automatically the moment a charge succeeds.

[Update your card](#)

Updating your card takes about a minute. And if something's holding you back, or the timing is just bad, reply to this email and tell me — I read every one and I'd rather sort it out with you than lose you over a card.

Best,
Michael
Founder, ZeroGEX

FROM ZeroGEX <hello@zerogex.io>

SUBJECT **Your ZeroGEX access ends September 19, 2026 — the first charge didn't go through**

VARIANT trial-conversion failure

Hello,

Your free trial has ended and the first subscription charge on your Visa card ending in 4242 still hasn't gone through. I've kept your full access switched on while the card is retried — but that runs out on September 19, 2026.

Stripe has made its final automatic attempt, so updating your card is the one thing that will pick this back up.

If nothing clears by then, the account simply moves to the free Public tier. Nothing is deleted — your account, your settings and your history all stay exactly as they are, and full access switches back on automatically the moment a charge succeeds.

Update your card

Updating your card takes about a minute. And if something's holding you back, or the timing is just bad, reply to this email and tell me — I read every one and I'd rather sort it out with you than lose you over a card.

Best,
Michael
Founder, ZeroGEX

Payment recovered

WHEN Seconds after a failed payment finally clears.

FIRED BY	subscription sync past_due → active → maybeHandlePaymentRecovery
CADENCE	Once per failure → recovery cycle
WHO GETS IT	payment_recovery_pending = 1, CAS-consumed on the sync back to active.
SENT-ONCE GUARD	payment_recovery_pending flag (CAS 1 → 0)
OPT-OUT	None (transactional)
SOURCE	core/mailer.ts:1996 · app/api/webhooks/stripe/route.ts:1083
NOTES	Only fires after a real failure — never on an ordinary renewal.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **You're all set — your ZeroGEX payment went through**

Hello,

Good news — the subscription payment we had trouble with earlier just went through, so your ZeroGEX account is fully active again. There's nothing more you need to do.

If your access was interrupted while the charge sorted itself out, it's all restored now, and you can jump straight back into the data.

[Open the live dashboard](#)

Sorry for the small bump. If you have any questions about the charge, just reply to this email — I'm happy to help.

Best,
Michael
Founder, ZeroGEX



ZeroGEX is a Folds of Honor Proud Supporter.

3% of every subscription funds educational scholarships for the families of fallen and disabled U.S. service members. [See /giving](#) for the running total.

Card expiring

WHEN When the card on file expires within 45 days and has not already been warned this calendar month. Checked daily.

FIRE BY	zerogex-web-card-expiry.timer → make card-expiry-reminders YES=1
CADENCE	Timer: daily 04:20 UTC (+0–10m jitter)
WHO GETS IT	subscription_status='active' AND stripe_subscription_id IS NOT NULL AND stripe_customer_id IS NOT NULL AND deleted_at IS NULL, then card expiry resolved live from Stripe
SENT-ONCE GUARD	card_expiry_notified_ym — year-month stamp, so it can warn again in a later month
OPT-OUT	None (billing)
SOURCE	core/mailer.ts:1390 · scripts/send-card-expiry-reminders.mts (THRESHOLD 45d, LIMIT 500/run)

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Heads-up: your card on file expires soon**

Hello,

Just a friendly heads-up: your Visa card ending in **4242** — the one on file for your ZeroGEX subscription — expires **10/2026**. Once it lapses, your next renewal could be declined and your access would pause until you update it.

Updating it takes about a minute from your account page and keeps everything running without a hitch — nothing else changes.

[Update your card](#)

If you've already updated your card (or your bank has issued a replacement Stripe can pick up automatically), you can ignore this. Any questions, just reply — I'm happy to help.

Best,
Michael
Founder, ZeroGEX

Referral reward

WHEN At the moment the referral is credited (the referee's subscription qualifies).

FIRED BY	core/referrals.ts — referral ledger pending → rewarded
CADENCE	Once per referral (UNIQUE referee_user_id)
WHO GETS IT	Referrer of a qualifying new subscription.
SENT-ONCE GUARD	referrals ledger row state
OPT-OUT	None (account credit notice)
SOURCE	core/mailer.ts:323 · core/referrals.ts:354

FROM ZeroGEX <hello@zerogex.io>
SUBJECT 🎉 You earned a free month on ZeroGEX
VARIANT credited

You earned a free month

Someone you referred just subscribed to ZeroGEX. We've added \$59.00 of account credit, which will be applied automatically to your next invoice.

[View your referrals](#)

Thanks for spreading the word.



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FROM ZeroGEX <hello@zerogex.io>
SUBJECT 🎉 You earned a free month on ZeroGEX
VARIANT banked

You earned a free month

Someone you referred just subscribed to ZeroGEX. You've earned a free month — we'll apply it as account credit automatically the next time you subscribe.

[View your referrals](#)

Thanks for spreading the word.



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CONVERSION NUDGES · 5 EMAILS

Signed up, never converted. All one-shot, all latched.

SYSTEMD TIMER MEMBER INBOX

Verify reminder

WHEN 2–168 hours after registering, if the address was never confirmed. Swept every 2 hours.

FIRED BY	zerogex-web-verify-reminders.timer → make verify-reminders YES=1
CADENCE	Timer: 00/2:20 UTC (every 2h, +0–5m jitter)
WHO GETS IT	tier='public' AND email_verified_at IS NULL AND stripe_subscription_id IS NULL AND verify_reminder_email_sent_at IS NULL AND deleted_at IS NULL AND created_at within [now-168h, now-2h]
SENT-ONCE GUARD	verify_reminder_email_sent_at — permanent one-shot
OPT-OUT	None (account-completion, not marketing)
SOURCE	core/mailer.ts:2810 · scripts/send-verify-reminders.mts (LAG 2h / LOOKBACK 7d)
NOTES	Mints a fresh 24h verify link at send time so the link in the mail is never already dead.

FROM

ZeroGEX <hello@zerogex.io>

SUBJECT

Finish setting up your ZeroGEX account

Hello,

I'm Michael, the founder of ZeroGEX. You created an account but the email address was never confirmed, so the account is still only half-set-up — and right now that's the one thing standing between you and the product.

Confirming your email unlocks everything, including the 7-day free trial: 7 days of full access, card on file but not charged until day 8, a heads-up email 48 hours before the first payment, and one-click cancel any time inside the trial.

Confirm email & unlock the trial

Or copy this URL into your browser:
<https://zerogex.io/verify?token=abc123def456>

This link expires in 24 hours.

If you didn't create a ZeroGEX account, you can safely ignore this — nothing further will happen. And if something went wrong or you have a question, just hit reply; I read every message myself.

Best,
Michael
Founder, ZeroGEX

Abandoned-checkout recovery

WHEN 24–168 hours after a billing_checkout_started audit row with no subscription to show for it. Swept every 6 hours.

FIRE BY	zerogex-web-checkout-recovery.timer → make checkout-recovery YES=1
CADENCE	Timer: 00/6:45 UTC (every 6h, +0–5m jitter)
WHO GETS IT	audit type='billing_checkout_started' in [now-168h, now-24h] AND stripe_subscription_id IS NULL AND checkout_recovery_email_sent_at IS NULL AND email_verified_at IS NOT NULL AND tier != 'admin' AND deleted_at IS NULL
SENT-ONCE GUARD	checkout_recovery_email_sent_at — permanent one-shot
OPT-OUT	None wired
SOURCE	core/mailer.ts:1185 · scripts/send-checkout-recovery.mts (LAG 24h / LOOKBACK 7d)
NOTES	Subject has three variants; a live founding deadline outranks a live promo deadline, which outranks the plain version.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Pick up where you left off at ZeroGEX**
VARIANT plain

Hello,

I noticed you started a ZeroGEX subscription recently but didn't finish. No pressure either way — sometimes a tab just gets closed.

If you'd like to pick it back up, the same plan is one click away:

Resume checkout

If ZeroGEX isn't the right fit, just ignore this — you won't hear from me again about it. And if anything stopped you from finishing, feel free to reply to this email. I read every message.

Best,
Michael
Founder, ZeroGEX

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Your ZeroGEX founding rate is still available — only until October 1, 2026**
VARIANT founding deadline live

Hello,

I noticed you started a ZeroGEX subscription recently but didn't finish. No pressure either way — but as a founding member you're still eligible for the locked-in founding rate, and that offer closes **October 1, 2026**.

If you'd like to pick it back up, the same plan is one click away:

Lock in the founding rate

After the deadline the founding rate is gone for good, so I wanted to give you a heads-up rather than let it lapse quietly. If ZeroGEX isn't the right fit, just ignore this — you won't hear from me again about it.

Best,
Michael
Founder, ZeroGEX

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Your ZeroGEX limited-time offer is still open — only until October 1, 2026**
VARIANT promo deadline live

Hello,

I noticed you started a ZeroGEX subscription recently but didn't finish. No pressure either way — but our **limited-time introductory pricing** is still live and closes **October 1, 2026**. Basic starts at **\$19.00/mo** and Pro at **\$39.00/mo**, with discounted annual plans too.

If you'd like to pick it back up, the same plan is one click away:

Claim the intro rate

Once the offer closes the standard rates come back, so I wanted to give you a heads-up rather than let it lapse quietly. If ZeroGEX isn't the right fit, just ignore this — you won't hear from me again about it.

Best,
Michael
Founder, ZeroGEX

Verified, never paid

WHEN 2–168 hours after verifying, if checkout was never opened. Swept every 2 hours.

FIRE BY	zerogex-web-verified-never-paid.timer → make verified-never-paid YES=1
CADENCE	Timer: 00/2:30 UTC (every 2h, +0–5m jitter)
WHO GETS IT	tier='public' AND email_verified_at IS NOT NULL AND subscription_lapsed != 1 AND stripe_subscription_id IS NULL AND verified_never_paid_email_sent_at IS NULL AND deleted_at IS NULL AND created_at within [now-168h, now-2h]
SENT-ONCE GUARD	verified_never_paid_email_sent_at — permanent one-shot
OPT-OUT	None wired
SOURCE	core/mailer.ts:2557 · scripts/send-verified-never-paid.mts
NOTES	Pitches the standard 7-day trial. No discount.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Quick note from Michael at ZeroGEX**

Hello,

I'm Michael, the founder of ZeroGEX. I noticed you signed up for an account but haven't tried the full product yet — wanted to reach out personally rather than route you through a generic marketing flow.

If you've been weighing the one week free trial: it's 7 days of full access, your card is on file but won't be charged until day 8, and we send a heads-up email 48 hours before the first payment so the conversion is never a surprise. If within the 7 days you find it is not the right fit, you can cancel in one click on the billing portal and you won't be charged.

If you have a question, a hesitation, or feedback on what's missing — just hit reply. I read every message myself, and customer notes are a big part of how I decide what to build next.

Start the free trial

Either way, thanks for giving ZeroGEX a look.

Best,
Michael
Founder, ZeroGEX

Reactivation (extended trial)

WHEN 21+ days after signing up, for a verified account that never started a trial. Checked daily. Max 50 per run.

FIRE BY	zerogex-web-reactivation.timer → make reactivation YES=1
CADENCE	Timer: daily 16:40 UTC (+0–5m jitter)
WHO GETS IT	tier='public' AND email_verified_at IS NOT NULL AND subscription_lapsed != 1 AND stripe_subscription_id IS NULL AND reactivation_email_sent_at IS NULL AND marketing_unsubscribed_at IS NULL AND created_at older than 21d
SENT-ONCE GUARD	reactivation_email_sent_at — permanent one-shot
OPT-OUT	Honors marketing_unsubscribed_at; RFC 8058 List-Unsubscribe + visible footer
SOURCE	core/mailer.ts:2705 · scripts/send-reactivation.mts (LAG 21d, LIMIT 50)
NOTES	The offer is a LONGER trial (30 days, granted server-side at ?reactivate=1), never a discount. This one DOES auto-send — unlike win-back and return-intent.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **I extended your ZeroGEX free trial to 30 days**
HEADERS List-Unsubscribe, List-Unsubscribe-Post

Hello,

It's Michael, the founder of ZeroGEX. A little while back you created an account but never started the trial — no worries, life gets busy. I'm reaching out one more time because I don't think 7 days was a fair test, and I'd rather fix that than lose you over it.

So I've set your trial to a full **30 days**.

Same full access — every gamma level, the daily Read that tells you in plain English whether SPX is pinned, squeezing, or set to break before the open, the live flow, the backtester, and the TradeWorkz™ bots with their fully public trade audit — just a lot more room to see whether it earns a place in your routine.

How the trial works, so nothing catches you off guard:

- **30 days** of full access, starting the moment you activate.
- Your card is on file but is **not charged until day 31**.
- We email you **48 hours before** that first charge — it is never a surprise.
- If it's not for you, **one click** in the billing portal cancels it and you won't be charged a cent.

[Start my 30-day trial](#)

And if something specific held you back — a missing feature, a question, price, or it just wasn't the right week — hit reply and tell me. I read every message myself, and it genuinely shapes what I build next.

Either way, thanks for giving ZeroGEX a look.

Best,
Michael
Founder, ZeroGEX

You're receiving this because you created a ZeroGEX account. [Unsubscribe from these emails.](#)

Founding final call

WHEN Never automatically. Run by hand the day before the founding deadline.

Does not send on its own — no timer is wired to it.

FIRED BY	make founding-final-call — no timer
CADENCE	Manual only
WHO GETS IT	Operator-selected cohort for the founding deadline.
SENT-ONCE GUARD	founding_final_call_email_sent_at — permanent one-shot
OPT-OUT	n/a
SOURCE	core/mailer.ts:1043 · scripts/send-founding-final-call.mts
NOTES	Included for completeness: Makefile target exists, no systemd timer does.

FROM ZeroGEX <hello@zerogex.io>

SUBJECT **Final reminder: ZeroGEX founding rate closes tomorrow, October 1, 2026**

Hello,

Just a final reminder that the ZeroGEX founding rate closes tomorrow, **October 1, 2026**.

I wanted to send one last note because you were part of the early ZeroGEX cohort, and I do not want anyone who intended to lock in the founding rate to miss the deadline.

Here's the offer:

- **Basic:** \$12.00/mo for the first year, normally \$29.00/mo
- **Pro:** \$19.00/mo for the first year, normally \$59.00/mo
- After the first year, you keep **25% off standard pricing** for as long as your subscription stays active
- **No charge today.** Your card will not be billed until October 8, 2026.

After **October 1, 2026**, founding access ends and future access will be at standard pricing.

If ZeroGEX has been useful to you, this is the best pricing I expect to offer, and it is my way of saying thank you to the people who were here early.

You can lock it in here:

Lock in my founding rate

If ZeroGEX is not for you right now, no worries at all. You can ignore this email and you won't hear from me again about the founding offer.

Thanks again for being part of the early ZeroGEX group. I genuinely appreciate the support, feedback, and encouragement many of you have shared.

Best,
Michael
Founder, ZeroGEX

RETENTION & CHURN · 3 EMAILS

Cancelled or lapsed. Two of the three do NOT auto-send — see the flag.

STRIPE WEBHOOK

MEMBER INBOX

Cancellation acknowledgment

WHEN Seconds after the member clicks Cancel — while they still have access through period end.

FIRED BY	subscription sync → cancel_at_period_end false → true → maybeHandleCancelAckTransition
CADENCE	Once per cancellation (re-arms if they reactivate then cancel again)
WHO GETS IT	cancel_at_period_end flipped to true.
SENT-ONCE GUARD	cancel_ack_email_sent_at — cleared on reactivation
OPT-OUT	None (transactional)
SOURCE	core/mailer.ts:2073 · app/api/webhooks/stripe/route.ts:1006
NOTES	Carries the signed one-click /save link — 25% off for a year, self-serve, no operator.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Sorry to see you go — mind sharing why?**
VARIANT with save link

Hello,

I saw you just canceled your ZeroGEX subscription — first, thank you. You've been a real part of what I've been building here, and I don't take that lightly.

You still have full access until **October 14, 2026**, so nothing changes yet on your end. I just wanted to reach out personally before that day comes.

If it comes down to price — keep your access at **25% off for a full year**, in one click. No re-subscribe, no re-entering a card.

Keep my access & claim 25% off

If you have a minute, I'd love to know what made you cancel. Even one sentence back on this email helps me a lot — I read every reply. Common ones I hear:

- The data wasn't what I expected
- Price is too high for how I trade
- I'm not trading as much right now
- Something specific was missing or broken
- Just trying it out for a stretch

Whatever the reason, I'd genuinely like to hear it.

Either way — thanks for giving ZeroGEX a shot. If you ever come back, your account will be here waiting.

Best,
Michael
Founder, ZeroGEX

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Sorry to see you go — mind sharing why?**
VARIANT conversion charge pending

Hello,

I saw you just canceled your ZeroGEX subscription — first, thank you. You've been a real part of what I've been building here, and I don't take that lightly.

First, so it doesn't catch you out: your trial had already ended when you canceled, so the payment for this billing period was already in motion and will go through shortly. **That is your final charge** — nothing renews after it — and you keep full access until **October 14, 2026**.

If it comes down to price — keep your access at **25% off for a full year**, in one click. No re-subscribe, no re-entering a card.

Keep my access & claim 25% off

If you have a minute, I'd love to know what made you cancel. Even one sentence back on this email helps me a lot — I read every reply. Common ones I hear:

- The data wasn't what I expected
- Price is too high for how I trade
- I'm not trading as much right now
- Something specific was missing or broken
- Just trying it out for a stretch

Whatever the reason, I'd genuinely like to hear it.

Either way — thanks for giving ZeroGEX a shot. If you ever come back, your account will be here waiting.

Best,
Michael
Founder, ZeroGEX

Win-back

WHEN The timer runs weekly, Mondays 16:35 UTC — but it only mails the OPERATOR a review digest. No member is mailed until someone runs make winback YES=1 by hand.

Does not send on its own — the installed unit only produces an operator review digest.

FIRED BY	zerogex-web-winback.timer → make winback DIGEST=1
CADENCE	Timer: Mon 16:35 UTC (+0–5m jitter) — review only
WHO GETS IT	subscription_lapsed=1 AND stripe_subscription_id IS NULL AND email_verified_at IS NOT NULL AND tier != 'admin' AND winback_email_sent_at IS NULL AND deleted_at IS NULL AND lapsed 30+ days ago
SENT-ONCE GUARD	winback_email_sent_at — cleared on welcome-back
OPT-OUT	Honors marketing_unsubscribed_at; plain-language opt-out footer
SOURCE	core/mailer.ts:2353 · scripts/send-winback.mts (LAG 30d)
NOTES	HUMAN IN THE LOOP. The installed unit passes DIGEST=1, not YES=1.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **A lot has changed at ZeroGEX — and your discount's ready**
VARIANT auto-apply discount

Hello,

It's been about a month since your ZeroGEX subscription ended, and I wanted to reach out personally — no hard sell, just a genuine note. I don't like losing people, and I'd rather hear from you than not.

A fair amount has changed since you left. A few of the bigger ones:

- **Trade Bias** — One signed directional call that fuses the gamma and volatility regime with live flow, tape and momentum.
- **Session rewind** — Replay any session on the gamma chart and watch how a level actually held.
- **TradeWorkz bots** — Automated strategies with a fully public trade audit — every entry and exit on the record.

I'll be honest: if you still trade the way you used to, I think a couple of these would genuinely change your workflow, and it's a little bit of a shame to be missing them.

And to make coming back easy, I've set aside **25% off for a year** for you — it's already on your account, so when you tap the button below you'll see the lower price before you confirm anything. No code to type, nothing to reply to.

Come back at a discount

No pressure at all, though. If the timing isn't right, just ignore this and I won't keep nudging you. But your account is still here exactly as you left it, the door's open, and I'd love to have you back.

If anything specific pushed you away — a missing feature, a bug, a pricing thing — just hit reply and tell me. I read every message myself, and it genuinely shapes what I build next.

Best,
Michael
Founder, ZeroGEX

You're receiving this because you created a ZeroGEX account. If you'd rather not hear from us, you can [delete your account](#) — that cancels any subscription and closes the account.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **A lot has changed at ZeroGEX since you left**
VARIANT manual (reply for discount)

Hello,

It's been about a month since your ZeroGEX subscription ended, and I wanted to reach out personally — no hard sell, just a genuine note. I don't like losing people, and I'd rather hear from you than not.

A fair amount has changed since you left. A few of the bigger ones:

- **Trade Bias** — One signed directional call that fuses the gamma and volatility regime with live flow, tape and momentum.
- **Session rewind** — Replay any session on the gamma chart and watch how a level actually held.
- **TradeWorkz bots** — Automated strategies with a fully public trade audit — every entry and exit on the record.

I'll be honest: if you still trade the way you used to, I think a couple of these would genuinely change your workflow, and it's a little bit of a shame to be missing them.

See what you've missed

No pressure at all, though. If the timing isn't right, just ignore this and I won't keep nudging you. But your account is still here exactly as you left it, the door's open, and I'd love to have you back.

If anything specific pushed you away — a missing feature, a bug, a pricing thing — just hit reply and tell me. I read every message myself, and it genuinely shapes what I build next.

Best,
Michael
Founder, ZeroGEX

You're receiving this because you created a ZeroGEX account. If you'd rather not hear from us, you can [delete your account](#) — that cancels any subscription and closes the account.

Return intent

WHEN The timer runs daily at 16:50 UTC — but it only mails the OPERATOR a review digest. No member is mailed until someone runs make return-intent YES=1 by hand.

Does not send on its own — the installed unit only produces an operator review digest.

FIRED BY	zerogex-web-return-intent.timer → make return-intent DIGEST=1
CADENCE	Timer: daily 16:50 UTC (+0–5m jitter) — review only
WHO GETS IT	Lapsed, verified, not deleted, not unsubscribed, non-admin, and a login_success AFTER their most recent stripe_subscription_deleted that is between 24h and 14 days old. Cooldown 90 days.
SENT-ONCE GUARD	return_intent_email_sent_at — a COOLDOWN anchor, not a one-shot: it re-arms
OPT-OUT	Honors marketing_unsubscribed_at; RFC 8058 List-Unsubscribe + visible footer
SOURCE	core/mailler.ts:3271 · core/returnIntent.ts (QUIET 24h / MAX_AGE 14d / COOLDOWN 90d)
NOTES	HUMAN IN THE LOOP. The only churn touch that fires on behaviour (they came back on their own) rather than a calendar. Carries no discount by design.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **What's changed at ZeroGEX since you left**
VARIANT price objection, 3 new
HEADERS List-Unsubscribe, List-Unsubscribe-Post

Hello,

I noticed your ZeroGEX account was active again recently, and I'd rather reach out myself than let you bump into a paywall and quietly leave.

Your account is exactly as you left it — the layouts, the symbols, the settings are all still on it. Nothing was deleted when the subscription ended, and resubscribing turns it all back on as it was.

You said the price was the problem, so I'll be straight about it: the monthly rate is what it is, but **annual billing** brings the effective monthly cost down a long way, and the side-by-side is right there on the pricing page. I'd rather have you on the plan that actually fits than not have you here at all.

Here's what's shipped since you left:

- **Trade Bias** — One signed directional call that fuses the gamma and volatility regime with live flow, tape and momentum.
- **Session rewind** — Replay any session on the gamma chart and watch how a level actually held.

- **TradeWorkz bots** — Automated strategies with a fully public trade audit — every entry and exit on the record.

Pick up where you left off

And if the answer is no, that's completely fine — no follow-up sequence, no countdown timer. You can reply to this and tell me what's missing, or ignore it entirely and I'll leave you alone.

Best,
Michael
Founder, ZeroGEX

You're receiving this because you have a ZeroGEX account. [Unsubscribe from emails like this.](#)

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Your ZeroGEX account is still here**
VARIANT neutral, nothing new
HEADERS List-Unsubscribe, List-Unsubscribe-Post

Hello,

I noticed your ZeroGEX account was active again recently, and I'd rather reach out myself than let you bump into a paywall and quietly leave.

Your account is exactly as you left it — the layouts, the symbols, the settings are all still on it. Nothing was deleted when the subscription ended, and resubscribing turns it all back on as it was.

In case it's useful, here's what we've been building:

Pick up where you left off

And if the answer is no, that's completely fine — no follow-up sequence, no countdown timer. You can reply to this and tell me what's missing, or ignore it entirely and I'll leave you alone.

Best,
Michael
Founder, ZeroGEX

You're receiving this because you have a ZeroGEX account. [Unsubscribe from emails like this.](#)

PRODUCT ALERTS · 1 EMAIL

Queue-driven, per-event.

QUEUE DRAIN MEMBER INBOX

TradeWorkz bot alert

WHEN Within ~60 seconds of the bot event. The timer drains the queue every minute.

FIRED BY	zerogex-web-tradeworkz-notify.timer → make tradeworkz-notify
CADENCE	Timer: every minute at :12s (+0–15s jitter)
WHO GETS IT	Queued rows in tw_notifications_log with channel='email' – one send per row.
SENT-ONCE GUARD	tw_notifications_log row state
OPT-OUT	Unfollow the bot from the dashboard
SOURCE	core/mailer.ts:2960 · scripts/tradeworkz-notify-deliver.mts
NOTES	The only high-frequency member email in the system. Volume tracks bot activity, not the member lifecycle.

FROMZeroGEX <hello@zerogex.io>

SUBJECT**Gamma Scalper closed LONG SPY +\$1.20K (+830.0%)**

VARIANTexit (win)

TRADEWORKZ™ EXITWIN

Gamma Scalper closed a LONG SPY trade

Structure	long call
Contracts	5
Entry	\$4.20
Exit	\$6.61
Realized P&L	+\$1.20K (+830.0%)
Reason	target hit

Open dashboard

You are receiving this because you followed Gamma Scalper on TradeWorkz™. Manage channels or unfollow from the bot card on the dashboard.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **Gamma Scalper opened SHORT SPY · 5 contracts @ \$4.20**
VARIANT entry

TRADEWORKZ™ ENTRY

Gamma Scalper opened a SHORT SPY trade

Structure	long put
Contracts	5
Entry	\$4.20
Target	\$6.50
Stop	\$3.10
Conviction	72%

Spot is above the gamma flip with a thick call wall at 660; dealers sell into strength here.

[Open dashboard](#)

You are receiving this because you followed Gamma Scalper on TradeWorkz™. Manage channels or unfollow from the bot card on the dashboard.

OPERATOR-ONLY · 7 EMAILS

Goes to the founder inbox. No member ever receives these.

SYSTEMD TIMER OPERATOR INBOX

Cancellation alert

WHEN Within 15 minutes of a churn event. Swept at :07, :22, :37, :52 past every hour.

FIRED BY	zerogex-web-cancellation-alerts.timer → make cancellation-alerts
CADENCE	Timer: every 15 min (+0–45s jitter)
WHO GETS IT	audit_events of type stripe_cancellation_requested (still has access) or stripe_subscription_deleted (access gone)
SENT-ONCE GUARD	cancellation_alert_sent audit row keyed to the churn event id
OPT-OUT	n/a — internal
SOURCE	core/mailer.ts:2383 · scripts/send-cancellation-alerts.mts
NOTES	Sends nothing to the member. Carries the Stripe cancellation survey verbatim plus pre-filled make commands.

FROMZeroGEX <hello@zerogex.io>

SUBJECT[ZeroGEX] member@example.com canceled — no reason given

CANCELED — STILL HAS ACCESS

member@example.com

No reason given · after 126 days (~4 months)

Access runs until Oct 14, 2026, 12:00 PM EDT — 29 days left to reply.

They typed nothing — only the survey reason above.

MEMBER

Member	member@example.com
Reason (survey)	No reason given
What they typed	(nothing)
Tier at churn	pro
Tenure	126 days (~4 months)
Signed up	May 12, 2026, 10:02 AM EDT
Canceled	Sep 15, 2026, 2:20 PM EDT
Access ends	Oct 14, 2026, 12:00 PM EDT

User id

user_123

NEXT

```
make diagnose-user EMAIL=member@example.com
```

```
make save-url EMAIL=member@example.com
```

```
make honor-winback-discount EMAIL=member@example.com DRY_RUN=1
```

No reason captured. Worth a short “what made you leave?” note; those replies are how the survey gets better.

[Admin dashboard](#) · sent by make cancellation-alerts

Win-back review digest

WHEN Weekly, Mondays 16:35 UTC.

FIRED BY	zerogex-web-winback.timer → make winback DIGEST=1
CADENCE	Timer: Mon 16:35 UTC (+0–5m jitter)
WHO GETS IT	One digest listing everyone the win-back email WOULD go to.
SENT-ONCE GUARD	n/a
OPT-OUT	n/a — internal
SOURCE	core/mailer.ts:2483 · scripts/send-winback.mts
NOTES	Embeds the real rendered member draft so the reviewer sees exactly what would ship.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT [ZeroGEX] Win-back review — 2 churned members ready (auto)

Win-back review — 2 eligible

Discount variant: **auto**. Nothing has been sent yet.

To send to everyone listed, run:

```
make winback YES=1
```

Recipients (2)

- member@example.com
- another@example.com

DRAFT PREVIEW — Subject: A lot has changed at ZeroGEX — and your discount's ready

Hello,

It's been about a month since your ZeroGEX subscription ended, and I wanted to reach out personally — no hard sell, just a genuine note. I don't like losing people, and I'd rather hear from you than not.

A fair amount has changed since you left. A few of the bigger ones:

- Trade Bias** — One signed directional call that fuses the gamma and volatility regime with live flow, tape and momentum.
- Session rewind** — Replay any session on the gamma chart and watch how a level actually held.

- **TradeWorkz bots** — Automated strategies with a fully public trade audit — every entry and exit on the record.

I'll be honest: if you still trade the way you used to, I think a couple of these would genuinely change your workflow, and it's a little bit of a shame to be missing them.

And to make coming back easy, I've set aside **25% off for a year** for you — it's already on your account, so when you tap the button below you'll see the lower price before you confirm anything. No code to type, nothing to reply to.

Come back at a discount

No pressure at all, though. If the timing isn't right, just ignore this and I won't keep nudging you. But your account is still here exactly as you left it, the door's open, and I'd love to have you back.

If anything specific pushed you away — a missing feature, a bug, a pricing thing — just hit reply and tell me. I read every message myself, and it genuinely shapes what I build next.

Best,
Michael
Founder, ZeroGEX

You're receiving this because you created a ZeroGEX account. If you'd rather not hear from us, you can [delete your account](#) — that cancels any subscription and closes the account.

Reactivation review digest

WHEN Never automatically. --digest is an exclusive branch that exits before sending, and the unit does not pass it.

Does not send on its own — no timer is wired to it.

FIRED BY	make reactivation DIGEST=1 — the installed timer passes YES=1, not DIGEST=1
CADENCE	Manual only
WHO GETS IT	One digest listing the cold signups in scope.
SENT-ONCE GUARD	n/a
OPT-OUT	n/a — internal
SOURCE	core/mailer.ts:2732 · scripts/send-reactivation.mts:446
NOTES	Asymmetry worth knowing: win-back and return-intent ship as digest-only, reactivation ships as live-send-only. Nobody reviews the reactivation batch before it goes out.

FROM ZeroGEX <hello@zerogex.io>

SUBJECT [ZeroGEX] Reactivation review — 2 cold signups ready (30-day trial)

Reactivation review — 2 eligible

Offer: extended **30-day** free trial. Nothing has been sent yet.

To send to everyone listed, run:

```
make reactivation YES=1
```

Recipients (2)

- member@example.com
- another@example.com

DRAFT PREVIEW — Subject: I extended your ZeroGEX free trial to 30 days

Hello,

It's Michael, the founder of ZeroGEX. A little while back you created an account but never started the trial — no worries, life gets busy. I'm reaching out one more time because I don't think 7 days was a fair test, and I'd rather fix that than lose you over it.

So I've set your trial to a full **30 days**.

Same full access — every gamma level, the daily Read that tells you in plain English whether SPX is pinned, squeezing, or set to break before the open, the live flow, the backtester, and the TradeWorkz™ bots with their fully public trade audit — just a lot more room to see whether it earns a place in your routine.

How the trial works, so nothing catches you off guard:

- **30 days** of full access, starting the moment you activate.
- Your card is on file but is **not charged until day 31**.
- We email you **48 hours before** that first charge — it is never a surprise.
- If it's not for you, **one click** in the billing portal cancels it and you won't be charged a cent.

Start my 30-day trial

And if something specific held you back — a missing feature, a question, price, or it just wasn't the right week — hit reply and tell me. I read every message myself, and it genuinely shapes what I build next.

Either way, thanks for giving ZeroGEX a look.

Best,
Michael
Founder, ZeroGEX

You're receiving this because you created a ZeroGEX account. [Unsubscribe from these emails.](#)

Return-intent review digest

WHEN Daily at 16:50 UTC.

FIRE BY	zerogex-web-return-intent.timer → make return-intent DIGEST=1
CADENCE	Timer: daily 16:50 UTC (+0–5m jitter)
WHO GETS IT	One digest per returning-member batch, with each member's angle.
SENT-ONCE GUARD	n/a
OPT-OUT	n/a — internal
SOURCE	core/mailer.ts:3300 · scripts/send-return-intent.mts

FROM ZeroGEX <hello@zerogex.io>
SUBJECT **[ZeroGEX] Return-intent review — 1 returning member ready**

Return-intent review — 1 eligible

These members churned, came back to the site, and have not been answered. Nothing has been sent yet. To send to everyone listed, run:

```
make return-intent YES=1
```

Recipients (1)

- member@example.com — left 2026-08-14, back 2026-09-15, angle: price

DRAFT PREVIEW — Subject: What's changed at ZeroGEX since you left

Hello,

I noticed your ZeroGEX account was active again recently, and I'd rather reach out myself than let you bump into a paywall and quietly leave.

Your account is exactly as you left it — the layouts, the symbols, the settings are all still on it. Nothing was deleted when the subscription ended, and resubscribing turns it all back on as it was.

You said the price was the problem, so I'll be straight about it: the monthly rate is what it is, but **annual billing** brings the effective monthly cost down a long way, and the side-by-side is right there on the pricing page. I'd rather have you on the plan that actually fits than not have you here at all.

Here's what's shipped since you left:

- **Trade Bias** — One signed directional call that fuses the gamma and volatility regime with live flow, tape and momentum.
- **Session rewind** — Replay any session on the gamma chart and watch how a level actually held.
- **TradeWorkz bots** — Automated strategies with a fully public trade audit — every entry and exit on the record.

Pick up where you left off

And if the answer is no, that's completely fine — no follow-up sequence, no countdown timer. You can reply to this and tell me what's missing, or ignore it entirely and I'll leave you alone.

Best,
Michael
Founder, ZeroGEX

You're receiving this because you have a ZeroGEX account. [Unsubscribe from emails like this.](#)

Scheduled-unit failure alert

WHEN

The instant any scheduled unit exits non-zero. Not a schedule — a failure hook.

FIRE BY	OnFailure=zerogex-web-alert@%n.service on every scheduled unit
CADENCE	Event-driven
WHO GETS IT	Any failing zerogex-web-* unit.
SENT-ONCE GUARD	None — every failure alerts
OPT-OUT	n/a — internal
SOURCE	scripts/send-unit-failure-alert.mts · deploy/systemd/zerogex-web-alert@.service
NOTES	This is the email that tells you the other emails stopped sending. Carries the last N journal lines. In the render below the journal excerpt is empty and systemctl errors — the capture ran outside systemd, so there was no journal to read. On the box those blocks carry the real failure output.

FROM

ZeroGEX <hello@zerogex.io>

SUBJECT

[ZeroGEX] ⚠ zerogex-web-trial-reminders.service FAILED

zerogex-web-trial-reminders.service

exited non-zero on vm at 2026-09-16T16:41:23.892Z.

(could not read: System has not been booted with systemd as init system (PID 1). Can't operate.

Failed to connect to bus: Host is down)

Last 40 journal lines:

-- No entries --

journalctl -u zerogex-web-trial-reminders.service -n 200 --no-pager
systemctl status zerogex-web-trial-reminders.service
systemctl start zerogex-web-trial-reminders.service

Signup-rate alarm

WHEN Hourly at :35 UTC. The script itself gates to active hours and only alerts when the signup rate drops below the floor.

FIRED BY	zerogex-web-signup-alarm.timer → make signup-alarm
CADENCE	Timer: hourly :35 (+0–3m jitter)
WHO GETS IT	Counts users.created_at over a rolling window.
SENT-ONCE GUARD	In-script rate gate
OPT-OUT	n/a — internal
SOURCE	scripts/send-signup-alarm.mts

FROM ZeroGEX <hello@zerogex.io>

SUBJECT [ZeroGEX] ⚠ Signup alarm — 0 new registration(s) in the last 5h

⚠ Signup alarm — registrations have flatlined

Only **0** new account(s) were created in the last **5h** (alarm floor: < 1). Checked at 12:41 ET. Last 24h: **0**.

**Registrations
per day (UTC),
last 7 days**

2026-09-14 **1**

2026-09-13 **1**

The signup *pipeline* may be fine while the *top of funnel* dried up — they're diagnosed differently. Fast triage:

1. **Acquisition?** Check your ad platform — are campaigns delivering & spending now, and is the click traffic real (not bots)?
2. **The form?** In fresh incognito with DevTools open, complete a signup at <https://zerogex.io/register> — watch for a console error on load, a failed `/api/auth/csrf`, or a non-200 `/api/auth/register`.
3. **The backend?** On the box: make `webhook-health` and make `status` (BFF online and on Node 22 — `node:sqlite` needs 22+).
4. **Just early?** The admin chart buckets by ET day; the trailing-window number above ignores the day boundary.

Open admin monitoring

Sent by `scripts/send-signup-alarm.mts` (systemd timer: hourly, active hours only, throttled). Tune the `SIGNUP_ALARM_*` vars in `frontend/.env.local`, or disable with `sudo systemctl disable --now zerogex-web-signup-alarm.timer`.

Folds of Honor donation reminder

WHEN Quarterly — the 5th of January, April, July and October at 09:00 ET.

FIREDBY	zerogex-web-foh-donation-reminder.timer → make foh-donation-reminder
CADENCE	Timer: 5th of Jan/Apr/Jul/Oct, 09:00 ET (+0–10m jitter)
WHO GETS IT	n/a — one reminder to the operator.
SENT-ONCE GUARD	n/a
OPT-OUT	n/a — internal
SOURCE	scripts/send-foh-donation-reminder.mts
NOTES	One of only two units pinned to America/New_York rather than server time.

FROM ZeroGEX <hello@zerogex.io>
SUBJECT [ZeroGEX] Q3 2026 closed — time to make the FOH donation

Q3 2026 just closed — time to make the FOH donation

Roughly 15 minutes end-to-end. Four steps below. Nothing to remember and no docs to consult — everything you need is in this email.

Step 1 — Compute 3% of the closing quarter's revenue

Paste this into **Stripe Sigma** (Dashboard → More → Sigma):

```
SELECT SUM(amount_paid) / 100.0 AS gross_usd_paid
FROM   invoices
WHERE  status = 'paid'
      AND billing_reason IN ('subscription_cycle', 'subscription_create')
      AND paid_at BETWEEN '2026-07-01' AND '2026-09-30';
```

Multiply the result by **0.03** to get the donation amount. Write it down.

Step 2 — Send the donation

Wire or use a card at the ZeroGEX-tracked partner page:

[Open the ZeroGEX FOH partner page](#)

Save the receipt PDF that FOH emails you afterward.

Step 3 — Publish on the site (one command)

On the EC2 box, from ~/zerogex-web:

```
git checkout release && git pull
make quarterly-receipt
```

Interactive. It'll ask for the amount, quarter, and date, then update content/giving/totals.json, commit, push, rebuild, and print the exact tweet for Step 4.

Suggested inputs when prompted:

- **Amount:** your number from Step 1 $\times$ 0.03
- **Quarter:** Q3 2026
- **Date:** today (default — press Enter)

Step 4 — Post the tweet

Step 3 prints the tweet text at the end. Copy it, open X ([@ZeroGEXOptions](#)), **attach the Folds of Honor Proud Supporter badge**, and post.

Badge download: <https://zerogex.io/folds-of-honor-proud-supporter.png>

Full mechanics: <https://zerogex.io/giving>. Detailed walkthrough if anything trips: docs/quarterly-receipt-workflow.md.

Sent by scripts/send-foh-donation-reminder.mts (systemd timer: 5th of Jan/Apr/Jul/Oct, 14:00 UTC).

Appendix — can send, but never on its own

These exist as Makefile targets or scripts. No timer, no webhook, no app path reaches them; a person runs them. Listed so the audit is provably complete.

SENDER	INVOKED BY	WHAT IT IS
Product-update newsletter	<code>make send-product-update</code>	Founder-voice broadcast; copy lives in docs/newsletters/2026-07-product-update.{html,txt}. Honors marketing_unsubscribed_at, invisible List-Unsubscribe header.
Founding final call	<code>make founding-final-call</code>	Deadline-day urgency closer. Latched per account.
Trial quickstart bridge	<code>make send-trial-quickstart</code>	One-shot backfill for members mid-trial before the new welcome shipped.
Welcome email (re-send)	<code>make send-welcome-email</code>	Operator re-send of the paid / founding / welcome-back email.
Payment-failed preview	<code>make payment-failed-preview</code>	Renders the dunning email to a chosen address. Testing aid.
Payment-recovered (manual)	<code>scripts/send-payment-recovered.mts</code>	Operator re-send of the recovery bookend.
Trial back-credit	<code>scripts/back-credit-trial.mts</code>	One-off: notified pre-trial monthly subscribers of a ~7/31-month credit. Inline HTML, not mailer.ts.
Founding July-1 delay credit	<code>scripts/credit-founders-july1-delay.mts</code>	One-off: notified founders of a one-month credit at the founding rate. Inline HTML, not mailer.ts.
API 403 incident notice	<code>make send-403-notice</code>	One-off incident notice to affected API users.
Quarterly Folds of Honor receipt	<code>make quarterly-receipt</code>	Interactive publish loop; optional EMAIL= mails the tweet draft to the operator.

Method. Every email body in this document is the real wire payload: the Resend transport was replaced with a capture stub and each sender invoked, so what you are reading is the exact html Resend would have received — not a mock-up of it. Timing came from the systemd units in `deploy/systemd/`, the Makefile targets they call, and the cohort SQL in each script. Nothing here was taken from the prose in `docs/automated-emails-audit.md`.

Two honest caveats about the renders. Sample values (names, card digits, amounts, dates, signed token links) are representative, not real — the copy around them is verbatim. And one report rule reaches inside an email body: `.z-email pre { white-space: pre-wrap }`, because a `<pre overflow-x:auto>` scrolls in a mail client but clips on paper. Everything else renders against browser defaults, as a mail client would give it.